

# Ryan Wilson

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## Education

University of Michigan College of Engineering – Ann Arbor, MI  
BSE - Computer Science

April 2024

## Work Experience

### Criteo

Ann Arbor, MI

Full Stack Software Engineer – Onsite Ads

July 2024 – January 2026

- Built C# .NET backend and Angular frontend for customer facing apps using Agile development processes
- Participated in on call, regularly solving SOS and business escalation tickets to prevent loss of revenue
- Led the creation of an API to supply data to a new app including setup and testing of the CI/CD pipeline
- Used behavior and test driven development when writing and completing tickets on Jira to participate in demos
- Worked on the systems design and implementation for a major refactor of the servability rules library to a new easily expandable API with complex rules and data structures interacting with custom entity framework

Backend Software Engineering Intern

May 2023 – April 2024

- Built .NET application using test-driven development and domain-driven design utilizing AWS SQS queues to improve latency on API calls by up to 14 seconds
- Led migration project from DynamoDB to in-house key-value store system resulting in annual savings of \$50k including multiple steps of communication of needs, documentation, .NET version management, and unit testing

### Quantum Signal AI

Saline, MI

Software Engineering Intern – Simulation and Sensors Team

May 2022 – August 2022

- Worked with Python, Docker, and ROS to streamline simulation usage and setup
- Developed startup script to reduce simulation startup from multiple commands spread between Docker containers to a single command decreasing startup time from 6 minutes to 2 minutes
- Used socket programming in a server client model to make CARLA simulation deterministic

## Technical Experience/Projects

### Self Directed

Unity Game Development – Holy Ship

January 2023 – Current

- Worked in a team of 2 to develop and publish a 2D local multiplayer party game on Steam using C# Unity

Full Size Skee-Ball Machine

December 2022 – January 2023

- Worked in a 3 person group to plan, budget, and build a fully functional Skee-Ball machine complete with electronics and sensors to record the users scores managed by a Raspberry Pi

### University of Michigan

Ann Arbor, MI

Michigan Mars Rover – Auton Team Member

September 2021 – September 2023

- Developed software with ROS on C++, and Python to help the rover with perception, navigation, and localization including the detection of AR tags, posing of the rover, and mapping of the rover's path

## Technical Skills

### Languages

- C/C++, C#, Python, Java, SQL Typescript, Javascript, HTML, CSS

### Development Tools

- Unix, Git, .NET, AWS Dynamo DB, Docker, Angular, ROS/ROS2, CARLA, Unity, Unreal Engine, MySQL

## Activities

### Speedcubing

April 2014 - Current

- Solving different types of Rubik's Cubes competitively with a personal best of 6.42 seconds on 3x3
- Competed in CubingUSA Nationals 2017. Ranked top 50 in the state of Michigan